

Anoushka Vyas

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SUMMARY

AI Engineer with 8+ months of experience building production-grade multi-agent systems, RAG pipelines, and LLM-driven backend architectures. Skilled across FastAPI, LangGraph, vector databases, and React/Next.js with strong expertise in retrieval systems, backend engineering, orchestration workflows, and scalable AI application development.

EDUCATION

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| ● K J Somaiya Institute of Technology | Dec 2021 - May 2025 |
| <i>Bachelor of Technology (B.Tech) in Information Technology - CGPA: 9.4/10</i> | <i>Mumbai, India</i> |

EXPERIENCE

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| ● Mphasis Ltd | Sep 2025 - Present |
| <i>Associate Software Engineer</i> | <i>Goregaon, Mumbai</i> |

- Designed and deployed a **production-grade multi-agent conversational AI system** using LangGraph, enabling parallel agent execution, conditional routing, and domain-specific reasoning workflows across 4+ specialised agent types currently handling production query routing.
- Built an **LLM-driven NLU pipeline** with multi-intent classification, structured entity extraction, and confidence-based fallback handling, reducing failures on ambiguous and edge-case queries.
- Implemented an **intelligent routing layer** mapping user intents to specialised agents with dynamic override logic and sub-task decomposition, replacing brittle rule-based flows.
- Developed a **hybrid RAG retrieval system** using reciprocal-rank fusion (dense + keyword retrieval), improving contextual grounding and reducing hallucinations in enterprise responses.
- Engineered a **PII redaction layer** for sensitive financial identifiers and credentials to ensure secure responses across all API surfaces.
- Designed a **conversation memory system** with context management and summarisation to support coherent multi-turn AI interactions.
- Built and instrumented a **FastAPI backend** with dependency injection, structured logging, tracing, and debugging endpoints for production observability.

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| ● Reliance Jio Infocomm | Jun 2024 - Aug 2024 |
| <i>Web Developer Intern</i> | <i>Ghansoli, Navi Mumbai</i> |

- Converted Figma designs into **responsive production-ready interfaces** using React and Bootstrap for internal enterprise tooling.
- Improved UI consistency and cross-device responsiveness through reusable component design and rendering optimisations.

SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, Java, SQL
- **AI / Generative AI:** Large Language Models (LLMs), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Hybrid Retrieval (RRF), Prompt Engineering, Semantic Search, LangChain, LangGraph, Semantic Kernel, Azure OpenAI, Gemini
- **Backend Technologies:** FastAPI, Node.js, Express.js, Spring Boot, REST APIs
- **Frontend Technologies:** React, Next.js, Angular, TailwindCSS
- **Databases:** PostgreSQL, MongoDB, MySQL, ChromaDB, Qdrant, Redis
- **Cloud and Tools:** Docker, Git, Postman, Azure OpenAI, Vercel

PROJECTS

- **CortexKitchen: Autonomous AI Operations Decision Engine**

Python / FastAPI / LangGraph / PostgreSQL / Qdrant / Redis / Prophet / Gemini 2.5 Flash / Next.js / Docker

- Designed and built a **9-node LangGraph orchestration system** with specialised agents for forecasting, reservations, complaint intelligence, inventory, menu analytics, aggregation, and critic evaluation using typed state and conditional routing.
- Implemented a **Decision Governance layer** where a Critic Agent evaluates AI-generated recommendations across operational feasibility, execution realism, scenario fit, and cost-benefit dimensions with persisted verdict reasoning.
- Built a **hybrid RAG + LLM architecture** combining PostgreSQL relational data with Qdrant vector memory across complaint memory, SOPs, and historical decision collections.
- Engineered a **Prophet-based forecasting pipeline** with daily and hourly demand predictions, confidence bounds, and peak-ratio extraction for downstream AI calibration.

- **DocQuery: Enterprise Knowledge Assistant**

LangChain / ChromaDB / Groq / Python / Streamlit

- Built a **multi-document RAG question-answering pipeline** with semantic chunking, ChromaDB vector retrieval, and grounded source-aware responses.
- Designed the end-to-end ingestion and retrieval workflow using LangChain with structured prompting and hallucination reduction techniques.

- **InspireAI: AI Content Generator**

Next.js / React / TypeScript / TailwindCSS / Gemini

- Built a **full-stack AI content generation platform** with Gemini API integration and real-time streaming generation.
- Developed a **responsive TypeScript and TailwindCSS interface** supporting dynamic prompt templates, customisation, and persistent generation history.